Reasoning Framework Optionality in VertiCore

Why LangGraph Is a Choice — Not a Dependency

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1. Purpose of This Document

This document explains an important architectural property of VertiCore:

> **VertiCore is not coupled to LangGraph.**

LangGraph is the *default* reasoning framework, but **any reasoning framework can be substituted** as long as it adheres to VertiCore's execution and safety contracts.

This design ensures long-term flexibility, avoids vendor lock-in, and allows the platform to evolve as reasoning frameworks improve.

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2. Core Architectural Principle

> **Reasoning is a pluggable component; execution is not.**

VertiCore treats reasoning engines as: - Stateless - Ephemeral - Replaceable

Temporal, policy enforcement, memory, and audit layers remain fixed.

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3. The Reasoning Contract (Non-Negotiable)

Any reasoning framework used inside VertiCore **must satisfy the following constraints**:

3.1 Execution Constraints

- Must run inside a single Temporal activity
- Must complete within bounded time
- Must be restartable without external state

3.2 State Constraints

- No persistent memory
- No implicit chat history
- All state passed in explicitly

3.3 Output Constraints

- Structured outputs only (JSON / schema)
- Explicit routing decisions
- No free-form side effects

3.4 Tool Constraints

- Cannot execute tools directly
- Must emit declarative tool intents

If a framework violates these rules, it cannot be safely integrated.

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4. Why LangGraph Fits Well

LangGraph is a strong default because it: - Encourages explicit graphs - Separates routing from execution - Supports bounded state - Is compatible with deterministic workflows

However, these are **properties**, not requirements.

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5. Alternative Reasoning Frameworks

VertiCore can support (or evolve toward):

5.1 Custom Rule-Based Engines

- Deterministic logic
- Regulatory-friendly
- Easy to audit

5.2 State Machines / DAG Engines

- BPMN-style reasoning
- Decision tables
- Hybrid AI + rules

5.3 Other LLM Reasoning Frameworks

- LangChain (restricted mode)
- DSPy-style programs
- Planner–executor models

As long as they respect the reasoning contract, they are viable.

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6. What VertiCore Explicitly Does NOT Allow

Regardless of framework choice, VertiCore forbids:

- Autonomous agent loops
- Cross-agent calls
- Hidden memory persistence
- Tool execution inside reasoning engines
- Time-based waiting inside reasoning

These restrictions are platform guarantees.

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7. Why This Matters Strategically

This optionality: - Prevents framework lock-in - Allows incremental evolution - Supports domain-specific reasoning engines - Protects long-term platform viability

VertiCore remains stable even as AI tooling changes.

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8. One-Sentence Summary

> **VertiCore standardizes execution, governance, and safety — not how reasoning is implemented.**

LangGraph is a powerful choice today, but the platform is designed to outlive any single reasoning framework.